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Expected utility approximation and portfolio optimisation

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ABSTRACT

Classical portfolio selection problems that optimise expected utility can usually not be solved in closed form. It is natural to approximate the utility function, and we investigate the accuracy of this approximation when using Taylor polynomials. In the important case of a Merton market and power utility we show analytically that increasing the order of the polynomial does not necessarily improve the approximation of the expected utility. The proofs use methods from the theory of parabolic second-order partial differential equations. All results are illustrated by numerical examples.

1. Introduction

Optimal control problems involving utility maximisation are an important tool in life and pension insurance. Two main types of application of this technique include questions of optimal dividend allocation, see for example Steffensen (2004) or indeed portfolio selection. The latter has a long pedigree as described below yet is also an active area of current research, see for example Donnelly et al. (2015) or Bernhardt and Donnelly (2019).

The purest form of such optimal consumption and portfolio selection problems is a classical topic in financial economics. Usually, Merton (1969) is credited with the first analysis of the time-consistent case. This problem is relevant not least because it has a closed-form solution that serves as a point of reference for more complicated settings.

Practically relevant situations where utility optimisation is employed may not have a closed-form solution so actuaries need approximation methods. The most basic such technique in utility maximisation appears to be the Taylor expansion of the utility function. To embed this in the historiography of optimal portfolio allocation, one can view the mean–variance approach as an approximation of expected utility formally obtained by truncating its Taylor series after two terms. Approximations are interesting and relevant for two main reasons. First, while a numerical analysis can shed light on particular scenarios (i.e. special cases) only, analytical methods allow to consider (and prove) important properties of the general case. This includes in particular the asymptotic behaviour of a solution when parameters become large or small. The second reason for using approximations lies in the challenges of numerically implementing optimisation schemes for example for dynamic portfolio management. Here an approximation of the value (and hence utility) function turns out to be key for establishing first-order conditions, see for example Brandt et al. (2005) and Cong and Oosterlee (2017).

This paper makes contributions in both directions as the techniques and results allow a better understanding of such approximations. To this end we analytically investigate the quality of the Taylor approximation of expected utility in the setting of Nordfang and Steffensen (2017). We thus consider an optimal portfolio selection problem in the context of a market consisting of a riskless asset whose price is denoted by B_t and a risky asset of price S_t whose dynamics follow a geometric Brownian motion. On the finite time horizon $[0, T]$, the investor operates a portfolio of total wealth X_t by varying the proportion ω invested in the risky asset. We assume that $\omega = \omega(t, x)$ is a deterministic function of time t and current wealth x. More precise assumptions are given in the subsequent section.

An optimal portfolio selection problem then seeks to find the maximum expected utility of terminal wealth and the corresponding strategy, i.e., the aim is to solve $\sup_\omega \left\{\mathbb{E}_{t,x}[u(X_T^\omega)]\right\}$, where we use the superscript ω to represent the dependence of a random variable, here terminal wealth, on the investment strategy. Using a power utility $u(x) = \frac{x^{1-\gamma}}{1-\gamma}$ with $\gamma > 0$ and $\gamma \neq 1$, the Taylor expansion to order n around a point x_0 becomes

$$u(x) = \sum_{i=0}^{n} c_i(\gamma) x_0^{1-\gamma-i}(x - x_0)^i + r_n(x),$$

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for a remainder term r_n and some constants $c_i(\gamma)$. A key idea of Nordfang and Steffensen (2017) is to choose $x_0 = \mathbb{E}_{t,x}[X_T^\omega]$, a function of t and x, leading to

$$\mathbb{E}_{t,x}[u(X_T^\omega)] = \sum_{i=0}^{n} c_i(\gamma)\mathbb{E}_{t,x}[X_T^\omega]^{1-\gamma-i}\mathbb{E}_{t,x}\left[\left(X_T^\omega - \mathbb{E}_{t,x}[X_T^\omega]\right)^i\right] + R_n^\omega(t,x), \tag{1}$$

where $R_n^\omega(t,x) = \mathbb{E}_{t,x}[r_n(X_T^\omega)]$.

Our main results are threefold.

(i) The decay of the remainder term $R_n^\omega(t,x)$ in x satisfies $R_n^\omega(t,x) \sim x^{1-\gamma}$ as $x \to \infty$ independently of n and ω. That means that for any strategy increasing the order n does not necessarily improve the accuracy of the approximation.

(ii) We prove analytically that for given t, x the absolute value of the remainder term $R_n^\omega(t,x)$ increases in n for n sufficiently large. This means that there is an optimal approximation order that minimises the remainder. This result explains numerical observations in the literature.

(iii) The third result concerns the investment strategy. Here, the strategy obtained for a Taylor-approximated utility function does not yield an asymptotically close approximation to the optimal strategy for $x \to \infty$.

Our analysis is done in the important case of the power utility function, which is part of the family of Constant Relative Risk Aversion (CRRA) utility functions. While this is predominant in actuarial applications, it already requires a fairly detailed presentation, which, however, enables the reader to adjust the approach to more complicated situations. The derivation of the results hinges on a careful analysis of properties of the probability density of the wealth process X_t. The key observation is that the singularity of (derivatives of) the power utility function at the origin are smoothened out by the rapid decay of the density function when taking expectations. Thus the analysis carries over to more general utility functions that exhibit a similar behaviour at the origin.

On a more general level, the optimisation problem (1) is of the form

$$\sup_{\omega} \left\{ f\left(t, x, y_1^\omega(t,x), \ldots, y_n^\omega(t,x)\right) \right\},$$

where $y_i^\omega(t,x) = \mathbb{E}_{t,x}[g_i(X_T^\omega)]$ for some functions g_i. These optimisation problems are time-inconsistent due to their dependence on the moments of terminal wealth where Bellman's principle of optimality does not apply. Thus the problem cannot be solved by classical Hamilton–Jacobi–Bellman (HJB) equations. Instead we use the idea of extended HJB equations given in Kryger et al. (2019) to find the optimal strategy ω^*.

We give a brief literature review relevant to Taylor expansions of expected utility. Markowitz (1952) is the first to use a mean–variance approximation in portfolio selection problems, i.e. a second-order Taylor approximation of expected utility. Samuelson (1970) shows that when risk is limited ($\sigma \to 0$), the mean–variance approximation is a good method to approximate expected utility and including higher-order terms does improve the approximation. Tsiang (1972) approximates CRRA utility functions by Taylor polynomials around $\mathbb{E}[X_T]$ and notes that the Taylor expansion is a function of the moments of the distribution of return, i.e.,

$$\mathbb{E}[u(X_T)] = \sum_{i=0}^{\infty} \frac{u^{(i)}\left(\mathbb{E}[X_T]\right)}{i!}\mathbb{E}\left[\left(X_T - \mathbb{E}[X_T]\right)^i\right] = \sum_{i=0}^{\infty} \frac{u^{(i)}\left(\mathbb{E}[X_T]\right)}{i!} m_i,$$

where m_i is the ith central moment of return X_T. The author comments that using only the first two terms of the Taylor expansion yields a useful approximation. Loistl (1976) shows that under power utility assumptions, the Taylor expansion of the utility function $u(X_T)$ around the mean $\mathbb{E}[X_T]$ converges only within the interval $\left[0, 2\mathbb{E}[X_T]\right]$. However, this interval of convergence only applies to the utility function itself and not to its expectation. The author also gives numerical examples with a log-normally distributed return X_T and a power utility function u. These examples show that the approximation of the expected utility becomes worse when adding higher-order terms. Levy and Markowitz (1979) extend the second-order Taylor expansion using the mean $\mathbb{E}[X_T]$ and two other points $\mathbb{E}[X_T] - \epsilon$ and $\mathbb{E}[X_T] + \epsilon$ for small $\epsilon > 0$ and conclude that the mean–variance analysis is more suitable when the probabilities of extreme returns are low. Pulley (1983) gives several numerical examples to support the mean–variance approximation of the logarithmic utility function in practice. In a related research avenue, several authors consider special cases to assess the accuracy of the mean–variance approximation without analytical proofs. Hlawitschka (1994) gives numerical examples by comparing the value of the true expected utility $\mathbb{E}[u(X_T)]$ and the two-moment Taylor approximation $u\left(\mathbb{E}[X_T]\right) + \frac{1}{2}u''\left(\mathbb{E}[X_T]\right)\mathrm{Var}[X_T]$ and claims that the usefulness of Taylor expansions is an empirical issue, and it is unnecessary to consider the convergence of the infinite series. The author reproduces the numerical examples given in Loistl (1976) and shows for the power utility that the mean–variance approximation does not change the priority order in which securities are chosen. Brockett and Garven (1998) calculate the remainder term of the Taylor expansion when $n = 3$ with a logarithmic utility and a log-normally distributed return with parameters (η, τ^2). The remainder term reads

$$\begin{aligned} R_3 &= \mathbb{E}[u(X_T)] - u\left(\mathbb{E}[X_T]\right) - \tfrac{1}{2}u''\left(\mathbb{E}[X_T]\right)m_2 - \tfrac{1}{6}u'''\left(\mathbb{E}[X_T]\right)m_3 \\ &= -\tfrac{1}{2}\tau^2 + \tfrac{1}{2}e^{\tau^2-1} - \tfrac{1}{3}\left(e^{3\tau^2} - 3e^{\tau^2} + 2\right), \end{aligned}$$

which is a function of τ. Here m_i is the ith central moment of return X_T. This is in support of Tsiang (1972), showing that the remainder term is a function of the variance of $\ln(X_T)$ and can be larger in magnitude than the truncated part. Brockett and Garven (1998) comment that Taylor polynomials may give a very poor approximation. Jondeau and Rockinger (2006) use the same form of Taylor expansion as we do but with an exponential utility. They show that including the third and the fourth terms of the Taylor expansion of expected utility improves the approximation in their case. Garlappi and Skoulakis (2011) show that one can improve the accuracy of Taylor approximations by a logarithmic transform as the distribution of transformed portfolio returns is more likely to be symmetric and of narrower support. It thus minimises the size of the deviations from the centre of the Taylor expansion and improves the approximation. The authors also give numerical examples for $\gamma > 1$ and show that for the lower-order of Taylor expansions, the approximations are poor. Skoulakis (2012) presents examples when the distribution of a logarithmic portfolio return is a Gram–Charlier distribution under the CRRA utility assumption and illustrates the advantage of using Taylor polynomials, confirming the findings of Hlawitschka (1994). The author draws two conclusions: first, that for a logarithmic return $\ln(X_T)$ the approximation does improve with the inclusion of higher-order terms of Taylor polynomials. Second, for X_T the approximation yields larger errors than for $\ln(X_T)$ and in some cases worsens with increasing order of the Taylor expansion. Finally, Nordfang and Steffensen (2017) suggest in numerical examples that higher-order Taylor expansions do not lead to significant benefits so that the second-order suffices.

This paper is organised as follows. Section 2 formulates the optimisation problem and recalls relevant techniques. In the subsequent Section 3, we describe the approximation method and analyse the remainder term using a power utility as an example. Section 4 summarises the main results.

2. Context and formulation of the optimisation problem

As Merton's Problem has been generalised in several different directions, this section illuminates the context of the portfolio optimisation problem considered in this paper. Of the many generalisations of Merton's Problem we sketch three that are particularly relevant for our approach as they introduce techniques to treat time-inconsistent situations where the optimisation problem does depend on current data (t,x) and the optimal strategy changes when time and wealth change. Starting from the original problem, we thus briefly describe recent generalisations by Björk and Murgoci (2010), Kronborg and Steffensen (2014) and Kryger et al. (2019).

2.1. Merton's problem

Merton (1969) considers the problem of optimal portfolio selection and consumption for an individual during a finite lifetime under a two-asset market assumption. The investor with wealth X_0 at initial time $t=0$ chooses a consumption strategy and divides their wealth between a riskless asset with rate of return r and a risky asset with drift μ and volatility σ. The dynamics of total wealth X_t read

$$dX_t = \Big(\big(r+\omega(t)\,(\mu-r)\big)X_t - C(t)\Big)dt + \omega(t)\sigma X_t dW_t,$$

where the investment strategy $\omega(t)$ and the consumption strategy $C(t)$ are functions of time t, and W_t is a Brownian motion.

The optimal control problem is then formulated in terms of the value function

$$V(t,x) = \sup_{\omega,C} \mathbb{E}_{t,x}\left[\int_t^T e^{-\rho s}u(C_s)ds\right],$$

where ρ is a discount rate. The corresponding HJB equation reads

$$\begin{aligned}
-\frac{\partial V(t,x)}{\partial t} &= \sup_{\omega,C}\Bigg\{ e^{-\rho t}u(C(t)) + \Big(\big(r+\omega(t)\,(\mu-r)\big)x - C(t)\Big) \\
&\qquad \times \frac{\partial V(t,x)}{\partial x} \\
&\qquad + \frac{1}{2}\sigma^2\omega(t)^2x^2\frac{\partial^2 V(t,x)}{\partial x^2}\Bigg\} \quad \text{and} \\
V(T,x) &= 0.
\end{aligned}$$

We denote the optimal investment strategy by $\omega^*(t)$ and optimal consumption strategy by $C^*(t)$. Under a CRRA utility

$$u(x) = \begin{cases} \frac{1}{1-\gamma}x^{1-\gamma} & \gamma \neq 1 \\ \ln x & \gamma = 1 \end{cases},$$

the optimal investment strategy $\omega^* = \frac{\mu - r}{\sigma^2\gamma}$ is constant, with Merton (1969) attributing this to the investor's attitude being independent of the wealth level.

2.2. Generalisation in Björk and Murgoci (2010)

Björk and Murgoci (2010) extend Merton's problem by assuming that there is a non-zero bequest valuation function F, a consumption function C and an additional function G (not necessarily with a specific economic interpretation) where at least one of these functions depends on (t,x). The authors then prove a verification theorem for an optimisation problem of the general form

$$\begin{aligned}
\sup_{\omega} \Bigg\{ &\mathbb{E}_{t,x}\left[\int_t^T C(t,x,s,X_s^{\omega},\omega(X_s^{\omega}))ds + F(t,x,X_T^{\omega})\right] \\
&+ G(t,x,\mathbb{E}_{t,x}[X_T^{\omega}])\Bigg\}.
\end{aligned}$$

This covers all previously known time-inconsistent examples. We give an example that is relevant to our work with the optimisation problem, the corresponding HJB equation and the optimal strategy.

Example 2.1. Consider a mean–variance utility problem of the form

$$\sup_{\omega}\left\{\mathbb{E}_{t,x}[X_T^{\omega}] - \frac{\gamma}{2}\mathrm{Var}_{t,x}[X_T^{\omega}]\right\},$$

with wealth dynamics

$$dX_t = \big(r+\omega(t,X_t)\,(\mu-r)\big)X_t dt + \omega(t,X_t)\sigma X_t dW_t.$$

Here $\gamma > 0$ is a constant that defines the risk aversion. The corresponding HJB equation becomes

$$\begin{aligned}
-\frac{\partial V(t,x)}{\partial t} &= \sup_{\omega}\Bigg\{\big(r+\omega(t,x)\,(\mu-r)\big)x\frac{\partial V(t,x)}{\partial x} \\
&\qquad + \frac{1}{2}\omega(t,x)^2\sigma^2x^2\frac{\partial^2 V(t,x)}{\partial x^2} \\
&\qquad - \frac{\gamma}{2}\omega(t,x)^2\sigma^2x^2\left(\frac{\partial g(t,x)}{\partial x}\right)^2\Bigg\}, \\
V(T,x) &= x,
\end{aligned}$$

where $g:[0,T]\times\mathbb{R}\to\mathbb{R}$ is a function that satisfies

$$\begin{aligned}
-\frac{\partial g(t,x)}{\partial t} &= \big(r+\omega^*(t,x)\,(\mu-r)\big)x\frac{\partial g(t,x)}{\partial x} \\
&\qquad + \frac{1}{2}\omega^*(t,x)^2\sigma^2x^2\frac{\partial^2 g(t,x)}{\partial x^2}, \\
g(T,x) &= x.
\end{aligned}$$

The optimal strategy is given by

$$\omega^*(t,x) = \frac{1}{x}\frac{\mu-r}{\sigma^2\gamma}e^{-r(T-t)},$$

which is inversely proportional to wealth.

2.3. Generalisation in Kronborg and Steffensen (2014)

In a related setting, Kronborg and Steffensen (2014) also include the consumption strategy and prove a verification theorem that optimises both investment and consumption. The optimisation problem reads

$$\sup_{c,\omega} f\left(t,x,y^{c,\omega}(t,x),z^{c,\omega}(t,x)\right),$$

where $f \in C^{1,2,2,2}\left([0,T]\times\mathbb{R}^3\right)$. For the verification theorem we refer to the cited paper and only present an example.

Example 2.2. Consider a mean–variance problem of the form

$$\begin{aligned}
\sup_{c,\omega}\Bigg\{&\mathbb{E}_{t,x}\left[\int_t^T e^{-\rho s}c(s)ds + e^{-\rho T}X_T^{c,\omega}\right] \\
&- \frac{\gamma}{2}\mathrm{Var}_{t,x}\left[\int_t^T e^{-\rho s}c(s)ds + e^{-\rho T}X_T^{c,\omega}\right]\Bigg\},
\end{aligned}$$

where $\gamma > 0$ is the risk aversion and $\rho \geq 0$ is a discount factor. The wealth dynamics read

$$dX_t = \Big(\big(r + \omega(t, X_t)(\mu - r)\big)X_t - c(t, X_t)\Big)dt + \omega(t, X_t)\sigma X_t dW_t,$$

and the admissible consumption strategies are assumed to be bounded, i.e., $c(s) \in [c_{min}, c_{max}]$ with $s \in [t, T]$. The optimal strategies are given by

$$c^*(t, x) = \begin{cases} c_{max} & \text{if } r < \rho \\ \text{non-defined} & \text{if } r = \rho \\ c_{min} & \text{if } r > \rho \end{cases}$$

$$\omega^*(t, x) = \frac{1}{x}\frac{\mu - r}{\sigma^2\gamma}e^{-(r-\rho)(T-t)}.$$

The optimal investment strategy ω^* for the case when $\rho = 0$ is the same as in Björk and Murgoci (2010). The discount factor ρ reflects how investors prefer to spend the wealth than to save it for the future. The optimal consumption strategy implies that investors with $\rho > r$ prefer to consume at the maximum level. They consider that the current wealth has higher purchase power and are impatient to save. On the opposite, investors with $\rho < r$ prefer savings to consumption.

2.4. Generalisation in Kryger et al. (2019)

Omitting the consumption in Sections 2.2 and 2.3, Kryger et al. (2019) consider a related problem and prove a verification theorem which we state here for completeness as it will be rephrased in our context later (Theorem 3.3).

Theorem 2.1 (*Theorem 1, Kryger et al. (2019)*)**.** *Consider the problem*

$$\sup_{\omega} f\left(t, x, y_1^\omega(t, x), \ldots, y_n^\omega(t, x)\right),$$

where functions $f \in C^{1,2,2,\ldots,2}\left([0, T] \times \mathbb{R}^{n+1}\right)$ and $g_1, \ldots, g_n \in C^2(\mathbb{R})$ are given, and we set $y_i^\omega(t, x) = \mathbb{E}_{t,x}[g_i(X_T^\omega)]$ for $i = 1, \ldots, n$. Assume the market dynamics to be

$$dX_t = \big(r + \omega(t, X_t)(\mu - r)\big)X_t dt + \omega(t, X_t)\sigma X_t dW_t.$$

Suppose there are functions $G_i^{\omega^*} : [0, T] \times \mathbb{R} \to \mathbb{R}$ satisfying

$$-\frac{\partial G_i^{\omega^*}(t, x)}{\partial t} = \big(r + \omega^*(t, x)(\mu - r)\big)x\frac{\partial G_i^{\omega^*}(t, x)}{\partial x} + \frac{1}{2}\sigma^2(\omega^*(t, x))^2x^2\frac{\partial^2 G_i^{\omega^*}(t, x)}{\partial x^2} \quad \text{and}$$

$$G_i^{\omega^*}(T, x) = g_i(x),$$

such that

$$\omega^*(t, x) = \arg\inf_{\omega}\Bigg\{-\frac{1}{2}\sigma^2(\omega(t, x))^2x^2\left(\sum_{i=1}^{n}\frac{\partial f(t, x)}{\partial y_i}\frac{\partial^2 G_i^{\omega^*}(t, x)}{\partial x^2}\right) - \big(r + \omega(t, x)(\mu - r)\big)x\left(\sum_{i=1}^{n}\frac{\partial f(t, x)}{\partial y_i}\frac{\partial G_i^{\omega^*}(t, x)}{\partial x}\right)\Bigg\} \tag{2}$$

is an admissible strategy with respect to $G_i^{\omega^*}$. Then the following assertions hold:

(i) The strategy $\omega^*(t, x)$ is the optimal strategy.
(ii) For the optimal strategy ω^* we have $y_i^{\omega^*}(t, x) = G_i^{\omega^*}(t, x)$.
(iii) The optimal value function is given by

$$V(t, x) = f\left(t, x, G_1^{\omega^*}(t, x), \ldots, G_n^{\omega^*}(t, x)\right).$$

Applying a first-order condition in (2) by formally differentiating with respect to ω, the optimal strategy is given in terms of f and $G_i^{\omega^*}$ as

$$\omega^*(t, x) = -\frac{\mu - r}{\sigma^2 x}\frac{\sum_{i=1}^{n}\frac{\partial f(t,x)}{\partial y_i}\frac{\partial G_i^{\omega^*}(t,x)}{\partial x}}{\sum_{i=1}^{n}\frac{\partial f(t,x)}{\partial y_i}\frac{\partial^2 G_i^{\omega^*}(t,x)}{\partial x^2}},$$

where we assume that $\sum_{i=1}^{n}\frac{\partial f(t,x)}{\partial y_i}\frac{\partial^2 G_i^{\omega^*}(t,x)}{\partial x^2} < 0$.

Kryger et al. (2019) give some economically-motivated examples in mean–variance style.

Example 2.3. We only present the value functions here.

(i) Generalised mean–variance optimisation:

$$\sup_{\omega}\left\{\rho(t, x)\mathbb{E}_{t,x}[X_T^\omega] - \frac{\gamma(t, x)}{2}\mathbb{E}_{t,x}\left[(X_T^\omega - k(t)\mathbb{E}_{t,x}[X_T^\omega])^2\right]\right\}$$

for a function k and with the restriction $\frac{\rho(t,x)}{\gamma(t,x)} = l_1(t)x + l_2(t)$ for some functions l_1 and l_2. This of course covers the classical mean–variance optimisation when we set $\rho = \gamma$ and $k \equiv 1$.

(ii) Generalised mean-standard-deviation optimisation:

$$\sup_{\omega}\left\{\mathbb{E}_{t,x}[X_T^\omega] - \gamma(t)\sqrt{\mathbb{E}_{t,x}\left[(X_T^\omega - k(t)\mathbb{E}_{t,x}[X_T^\omega])^2\right]}\right\}$$

for functions k and γ.

(iii) Scaled mean–variance optimisation:

$$\sup_{\omega}\left\{\rho(t, x)\mathbb{E}_{t,x}[X_T^\omega] - \gamma(t, x)\frac{\mathbb{E}_{t,x}\left[(X_T^\omega - \mathbb{E}_{t,x}[X_T^\omega])^2\right]}{\mathbb{E}_{t,x}[X_T^\omega]}\right\}$$

with the restriction $\frac{\rho(t,x)}{\gamma(t,x)} = q(t)$.

In general, these problems cannot be solved in closed form so we now consider analytical aspects of the approximation approach presented in Nordfang and Steffensen (2017).

3. The approximation method and statement of key results

In this section, we state the market assumptions and provide the analytic background to the approximation method in Nordfang and Steffensen (2017). The idea as in Nordfang and Steffensen (2017) is to expand the expectation $\mathbb{E}_{t,x}[u(X_T^\omega)]$ around the conditional expected point $\mathbb{E}_{t,x}[X_T^\omega]$ using Taylor polynomials. Our contribution is an analysis of the remainder term, and this is non-trivial since even if the remainder term of the utility function is small, taking its expectation can lead to an unexpected behaviour due to a "smearing" effect.

3.1. Assumptions on the market model and strategies

To model the financial market, we make the following assumption.

Assumption 3.1 (*Merton Market*)**.** We assume a two-asset financial market with dynamics $dB_t = rB_t dt$ and $dS_t = \mu S_t dt + \sigma S_t dW_t$, where B_t is a riskless asset with rate of return r and S_t is a risky asset with drift μ and volatility σ.

We also assume a self-financing portfolio $(1-\omega(t, X_t), \omega(t, X_t))$ whose total wealth X_t at time t satisfies the stochastic differential equation

$$dX_t = \big(r + \omega(t, X_t)(\mu - r)\big)X_t dt + \omega(t, X_t)\sigma X_t dW_t, \tag{3}$$

where $\omega(t, X_t)$, a function of time and wealth, is the proportion invested in the risky asset.

The next assumption concerns admissible strategies.

Assumption 3.2. We make the following assumptions on time, wealth and admissible strategies.

(i) We assume that time and wealth satisfy $(t, x) \in [0, T] \times \mathbb{R}_+$.
(ii) We assume that the strategy function $\omega(t, x)$ is bounded, i.e., for some constants λ_1 and λ_2, one has $0 < \lambda_1 \leq \omega(t, x) \leq \lambda_2$.
(iii) We assume that the strategy function $\omega(t, x)$ is Hölder continuous, i.e., for $(t_1, x_1), (t_2, x_2) \in [0, T] \times \mathbb{R}_+$, we assume

$$|\omega(t_1, x_1) - \omega(t_2, x_2)| \leq K \cdot \left(|x_1 - x_2|^\nu + |t_1 - t_2|^{\nu/2}\right)$$

for some constants $\nu \in (0, 1]$ and $K \in \mathbb{R}_+$.

These stipulations can be interpreted in economic terms.

(i) The wealth x is restricted to a positive value to avoid bankruptcy.
(ii) The two-sided bounds for strategies reflect a restriction on investment behaviour. Investors set limits $1 - \lambda_1$ for the riskless asset and λ_2 for the risky asset. Note that this is no restriction in practice, e.g. the strategy ω^* in Merton's problem is bounded from below if $\mu > r$.
(iii) The constant ν is not fixed and may be different for different strategies. The Hölder assumption is required for technical reasons to allow the argument via partial differential equations. No doubt, it can be relaxed at the cost of more sophisticated analytic methods. The main message of this paper would not suffer, however, the presentation would be much more involved.

3.2. Asymptotics of the remainder term

Recall that our optimisation problem reads $\sup_\omega \mathbb{E}_{t,x}[u(X_T^\omega)]$, where the superscript ω emphasises the dependence on the investment strategy. Suppose we are given a utility function $u \in C^\infty(\mathbb{R}_+)$. Then for $n \in \mathbb{N}_+$, its nth order Taylor expansion around the point x_0 with an integral remainder term reads

$$u(x) = \sum_{i=0}^{n} \frac{u^{(i)}(x_0)}{i!}(x - x_0)^i + \frac{(x - x_0)^{n+1}}{n!} \times \int_0^1 (1 - z)^n u^{(n+1)}\big(zx + (1 - z)x_0\big) dz.$$

Expanding $u(X_T^\omega)$ around $x_0 = E_{t,x}[X_T^\omega]$ and taking the expectation yields

$$\mathbb{E}_{t,x}\left[u(X_T^\omega)\right] = \sum_{i=0}^{n} \frac{u^{(i)}\left(\mathbb{E}_{t,x}[X_T^\omega]\right)}{i!} \mathbb{E}_{t,x}\left[\left(X_T^\omega - \mathbb{E}_{t,x}[X_T^\omega]\right)^i\right] + R_n^\omega(t, x) \tag{4}$$

with the remainder term

$$R_n^\omega(t, x) = \mathbb{E}_{t,x}\left[\int_0^1 \frac{(1 - z)^n}{n!}\left(X_T^\omega - \mathbb{E}_{t,x}\left[X_T^\omega\right]\right)^{n+1} \times u^{(n+1)}\left(zX_T^\omega + (1 - z)\mathbb{E}_{t,x}[X_T^\omega]\right) dz\right].$$

Inserting the power utility function $u(x) = \frac{1}{1-\gamma}x^{1-\gamma}$ for $\gamma > 0$ and $\gamma \neq 1$ into (4) yields

$$\begin{aligned}\mathbb{E}_{t,x}\left[u(X_T^\omega)\right] = &\binom{-\gamma}{n}\frac{(-1)^{n+1}}{-\gamma}\frac{1 + (1 - \gamma)(n - 1)}{1 - \gamma}\mathbb{E}_{t,x}[X_T^\omega]^{1-\gamma} \\ &+ \binom{-\gamma}{n}\sum_{i=2}^{n}\binom{n}{i}\frac{(-1)^{n+i}}{1 - \gamma - i}\mathbb{E}_{t,x}[X_T^\omega]^{1-\gamma-i} \\ &\times \mathbb{E}_{t,x}\left[\left(X_T^\omega\right)^i\right] \\ &+ R_n^\omega(t, x).\end{aligned}$$

Here, we set $\varphi^\omega(t, x) = \mathbb{E}_{t,x}[X_T^\omega]$.

To state our first main result, we recall the definition of two functions being asymptotically equivalent. Here and in the following we employ the usual O- and o-notation of Landau.

Definition 3.1 (*Erdélyi (1956), Chapter 1 Section 3*)**.** Two functions $f, g : \mathbb{R} \to \mathbb{R}$ are called *asymptotically equivalent*, denoted by $f \sim g$, if $f(x) - g(x) = o(g(x))$ as $x \to \infty$.

Our first result concerns the asymptotics of $R_n^\omega(t, x)$ in x.

Theorem 3.1. *Under Assumptions 3.1 and 3.2 the following assertions hold for any admissible strategy ω.*

(i) Given an admissible strategy ω, there is a function $M^\omega \in C^1([0, T])$ such that the remainder term $R_n^\omega(t, x)$ is asymptotically equivalent to $M^\omega(t) \cdot \varphi^\omega(t, x)^{1-\gamma}$, i.e., $R_n^\omega(t, x) \sim M^\omega(t) \cdot \varphi^\omega(t, x)^{1-\gamma}$.
(ii) Moreover, $R_n^\omega(t, x)$ is bounded by $x^{1-\gamma}$ and dominated by $x^{1-\gamma+\epsilon}$ for any $\epsilon > 0$, i.e., $R_n^\omega(t, x) = O(x^{1-\gamma})$ and $R_n^\omega(t, x) = o(x^{1-\gamma+\epsilon})$.

Note that the orders of $\varphi(t, x)$ and x in the asymptotics are independent of n.

See Appendix A.2 for the proof. The idea of the proof is to exploit the link between stochastic differential equations and partial differential equations as expressed by the Feynman–Kac formula. The key object is the probability density of X_t. Note that in stochastic analysis it is not unusual to work directly with the probability density, see Zhang (2005) for example.

Remark. We make two comments.

(i) Whenever one uses a Taylor approximation, one hopes that the decay of the remainder term in x increases with increasing order n. However, in this case, the remainders keep their growth/decay in x regardless of n. The reason is that while the approximation may be good for u pointwise, taking the expectation "smears" the good and bad approximations so that overall the growth/decay in x is constant. This suggests that a more sophisticated approximation technique is required that takes into account the weighting expressed by the probability density of X_t.
(ii) The point around which we take the Taylor expansion is taken as in Nordfang and Steffensen (2017), i.e. we expand the utility function around $\mathbb{E}_{t,x}[X_T^\omega]$, which is a function of t and x. The rationale for choosing $\mathbb{E}_{t,x}[X_T^\omega]$ is that the approximation with a dynamically adjusting should reflect more information available to the investor. The alternative – used by most of the cited contributions – would be to select a fixed point x_0, for which the natural choice would be $x_0 = \mathbb{E}[X_T^\omega]$. However, the numerical examples in Nordfang and Steffensen (2017) show that using this point gives a worse approximation.

Table 1
Sets of parameters.

Set	γ	μ	r	σ^2	ω^*
1	4	0.36	0.04	0.32	0.25
2	0.5	0.08	0.04	0.32	0.25
3	4	0.36	0.04	0.25	0.32

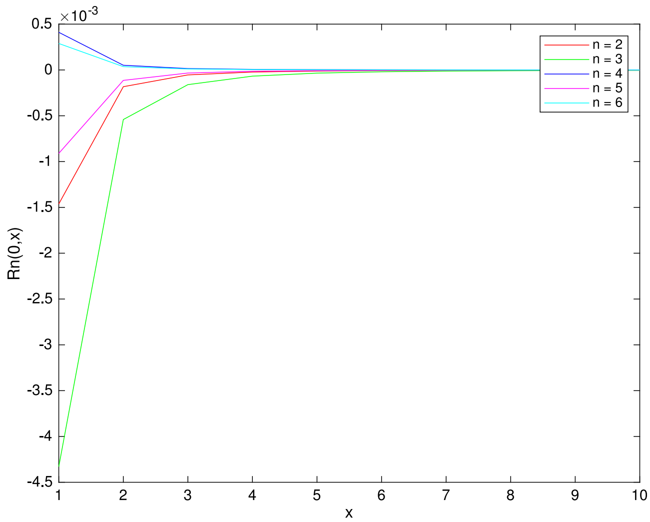

Fig. 1. Dependence of $R_n^\omega(0,x)$ on x with $n = 2, 3, 4, 5, 20$, Set 1.

Example 3.1. To illustrate the theorem we consider a simple case of where X_T has a lognormal distribution with explicit density function. This corresponds to Merton's problem where the optimal strategy is constant and given by $\omega^* = (\mu - r)/\sigma^2\gamma$. We calculate the remainder term numerically by computing the difference between the expected utility and the nth order Taylor approximation. The three scenarios (Sets 1–3) are determined in terms of parameters for (3) as given in Table 1 with initial and terminal time $t = 0$ and $T = 1$. The first set of parameters is from Nordfang and Steffensen (2017).

Figs. 1 and 2 nicely illustrate the assertions of Theorem 3.1: the remainder decays in x when $\gamma > 1$ and grows when $\gamma < 1$.

3.3. Growth of the remainder term in the approximation order

The pointwise dependence of the remainder term $R_n^\omega(t,x)$ on n can be formalised as follows.

Theorem 3.2. *Under Assumptions 3.1 and 3.2 the following assertion holds for any admissible strategy ω: Given t and x, there is an n_0 such that the absolute value of the remainder term $\left|R_n^\omega(t,x)\right|$ increases in n for $n \geq n_0$. This means that there is an optimal order that minimises the remainder term.*

The reader is referred to Appendix A.3 for the proof of this result.

Remark. This result confirms a numerical example (in the context of log-normally distributed returns) given in Loistl (1976) (Table 4 in *loc. cit.*) indicating that the approximation error decreases in the order for small n and then increases rapidly. See also Table 6 in Hlawitschka (1994) for a similar observation, however, based on empirical return distributions.

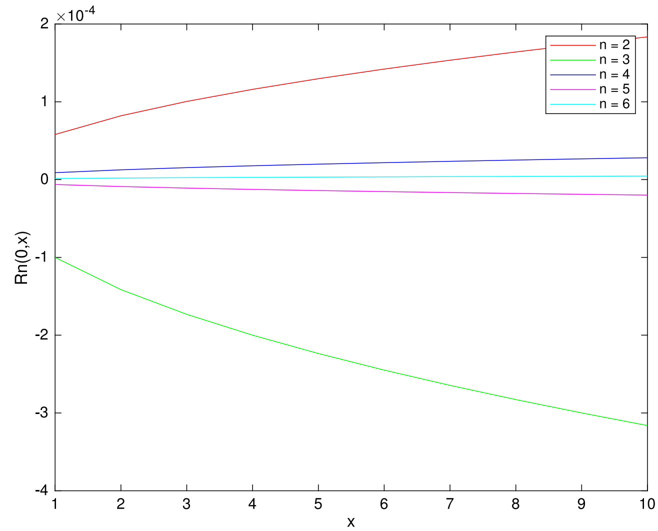

Fig. 2. Dependence of $R_n^\omega(0,x)$ on x with $n = 2, 3, 4, 5, 20$, Set 2.

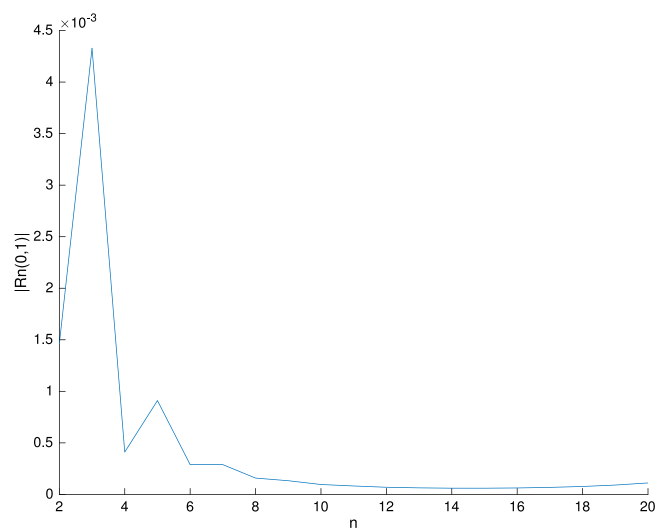

Fig. 3. Dependence of $\left|R_n^\omega(0,1)\right|$ on n, Set 1.

Example 3.2. In the setting of Example 3.1 we illustrate the dependence of $\left|R_n^\omega(t,x)\right|$ on n, with $x = 1$, $t = 0$ and $T = 1$. The results are shown in Figs. 3 and 4 referring to the parameter Set 1 and 3, respectively.

We make several observations here:

(i) The truncated Taylor polynomial of order three performs worse than the second-order, while the fourth-order polynomial is even better than the second-order. We clearly see the optimal approximation order beyond which the approximation becomes worse when n grows.
(ii) In both figures, the remainder terms with an order $n = 2$ are between 0.001 and 0.003, so that the mean–variance approximation to the expected utility appears suitable for practical applications.

3.4. The approximate optimal portfolio

We now analyse the effects of using Taylor expansions on portfolio selection. By Taylor-expanding the utility function, the

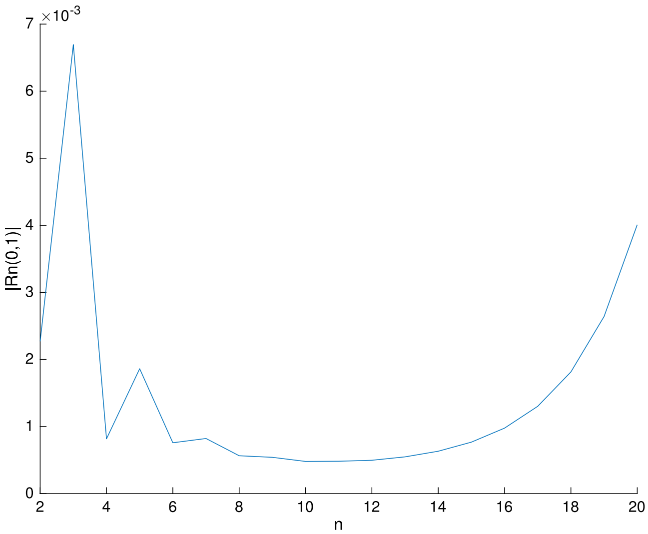

Fig. 4. Dependence of $\left|R_n^{\omega}(0,1)\right|$ on n, Set 3.

problem becomes

$$\sup_{\omega}\mathbb{E}_{t,x}\left[u\left(X_T^{\omega}\right)\right]=\sup_{\omega}\Bigg\{\binom{-\gamma}{n}\frac{(-1)^{n+1}}{-\gamma}\frac{1+(1-\gamma)(n-1)}{1-\gamma}\times\mathbb{E}_{t,x}[X_T^{\omega}]^{1-\gamma}+\binom{-\gamma}{n}\sum_{i=2}^{n}\binom{n}{i}\frac{(-1)^{n+i}}{1-\gamma-i}\mathbb{E}_{t,x}[X_T^{\omega}]^{1-\gamma-i}\times\mathbb{E}_{t,x}\left[\left(X_T^{\omega}\right)^i\right]+R_n^{\omega}(t,x)\Bigg\}.$$

Note that the optimisation problem can be expressed in the form of Theorem 2.1 through the value function

$$V(t,x)=\sup_{\omega}\left\{f\left(t,x,y_1^{\omega}(t,x),\ldots,y_n^{\omega}(t,x)\right)+R_n^{\omega}(t,x)\right\},$$

where $y_i^{\omega}(t,x)=\mathbb{E}_{t,x}[g_i(X_T^{\omega})]$ for suitable deterministic functions $g_i(X_T^{\omega})$.

We extend Theorem 2.1 to include a remainder term.

Theorem 3.3. *Consider the optimisation problem above under Assumption 3.1. Suppose there are functions $G_i^{\omega^*}(t,x)$ and $H_n^{\omega^*}(t,x)$ that satisfy*

$$-\frac{\partial G_i^{\omega^*}(t,x)}{\partial t}=\big(r+\omega^*(t,x)(\mu-r)\big)x\frac{\partial G_i^{\omega^*}(t,x)}{\partial x}+\frac{1}{2}\sigma^2\omega^*(t,x)^2x^2\frac{\partial^2 G_i^{\omega^*}(t,x)}{\partial x^2},$$

$$-\frac{\partial H_n^{\omega^*}(t,x)}{\partial t}=\big(r+\omega^*(t,x)(\mu-r)\big)x\frac{\partial H_n^{\omega^*}(t,x)}{\partial x}+\frac{1}{2}\sigma^2\omega^*(t,x)^2x^2\frac{\partial^2 H_n^{\omega^*}(t,x)}{\partial x^2},$$

$$G_i^{\omega^*}(T,x)=g_i(x)\quad\text{and}$$

$$H_n^{\omega^*}(T,x)=0,$$

where

$$\omega^*(t,x)=\arg\inf_{\omega}\Bigg\{-\frac{1}{2}\sigma^2(\omega(t,x))^2x^2\left(\sum_{i=1}^{n}\frac{\partial f(t,x)}{\partial y_i}\frac{\partial^2 G_i^{\omega^*}(t,x)}{\partial x^2}+\frac{\partial^2 H_n^{\omega^*}(t,x)}{\partial x^2}\right)-\big(r+\omega(t,x)(\mu-r)\big)x\left(\sum_{i=1}^{n}\frac{\partial f(t,x)}{\partial y_i}\frac{\partial G_i^{\omega^*}(t,x)}{\partial x}+\frac{\partial H_n^{\omega^*}(t,x)}{\partial x}\right)\Bigg\}.$$

Then we have the following assertions for the optimal strategy ω^* and the optimal value function V.

(i) The strategy ω^* is the optimal strategy.
(ii) We have $y_i^{\omega^*}(t,x)=G_i^{\omega^*}(t,x)$ and $R_n^{\omega^*}(t,x)=H_n^{\omega^*}(t,x)$.
(iii) Moreover, $V(t,x)=f\left(t,x,G_1^{\omega^*}(t,x),\ldots,G_n^{\omega^*}(t,x)\right)+H_n^{\omega^*}(t,x)$.

The terminal condition reads $H_n^{\omega^*}(T,x)=0$ because the remainder term satisfies $R_n^{\omega}(T,x)=0$. See Appendix A.4 for the proof of the theorem.

The optimal strategy $\omega^*(t,x)$ can be stated explicitly as

$$\omega^*(t,x)=-\frac{\mu-r}{\sigma^2x}\cdot\frac{\sum_{i=1}^{n}\frac{\partial f(t,x)}{\partial y_i}\frac{\partial G_i^{\omega^*}(t,x)}{\partial x}+\frac{\partial H_n^{\omega^*}(t,x)}{\partial x}}{\sum_{i=1}^{n}\frac{\partial f(t,x)}{\partial y_i}\frac{\partial^2 G_i^{\omega^*}(t,x)}{\partial x^2}+\frac{\partial^2 H_n^{\omega^*}(t,x)}{\partial x^2}},$$

where we assume that $\sum_{i=1}^{n}\frac{\partial f(t,x)}{\partial y_i}\frac{\partial^2 G_i^{\omega^*}(t,x)}{\partial x^2}+\frac{\partial^2 H_n^{\omega^*}(t,x)}{\partial x^2}<0$. On the other hand, according to Nordfang and Steffensen (2017) the approximate strategy is

$$\hat{\omega}^*(t,x)=-\frac{\mu-r}{\sigma^2x}\cdot\frac{\sum_{i=1}^{n}\frac{\partial f(t,x)}{\partial y_i}\frac{\partial G_i^{\omega^*}(t,x)}{\partial x}}{\sum_{i=1}^{n}\frac{\partial f(t,x)}{\partial y_i}\frac{\partial^2 G_i^{\omega^*}(t,x)}{\partial x^2}},$$

in the notation of the present paper. The approximation error is given by $\omega^*-\hat{\omega}^*$, and it is desirable that this error be small. In the framework of asymptotic analysis, a good notion of "small" would be that as x becomes large, the derivatives of $H_n^{\omega^*}$ decay faster than the terms involving the derivatives of f and $G_i^{\omega^*}$. This is, however, not the case.

The H-derivatives play the role of remainder terms. For the optimal strategy these can be given explicitly (see (A.9) and (A.10) in the Appendix), and we denote them by $\frac{\partial H_n^{\omega^*}(t,x)}{\partial x}=R_{1,n}^{\omega^*}(t,x)$ and $\frac{\partial^2 H_n^{\omega^*}(t,x)}{\partial x^2}=R_{2,n}^{\omega^*}(t,x)$. These remainders have the following asymptotic growth properties.

Theorem 3.4. *Under Assumptions 3.1 and 3.2 and for any admissible strategy ω the following assertions hold.*

(i) The terms $R_{1,n}^{\omega}(t,x)$ and $R_{2,n}^{\omega}(t,x)$ are bounded by $x^{-\gamma}$ and $x^{-\gamma-1}$ asymptotically and for any $\epsilon>0$, they are dominated asymptotically by $x^{-\gamma+\epsilon}$ and $x^{-\gamma-1+\epsilon}$, respectively. That is $R_{1,n}^{\omega}(t,x)=O(x^{-\gamma})$, $R_{1,n}^{\omega}(t,x)=o(x^{-\gamma+\epsilon})$, and $R_{2,n}^{\omega}(t,x)=O(x^{-\gamma-1})$, $R_{2,n}^{\omega}(t,x)=o(x^{-\gamma+\epsilon-1})$.
(ii) The term $\sum_{i=1}^{n}\frac{\partial f^{\omega}(t,x)}{\partial y_i}\frac{\partial y_i^{\omega}(t,x)}{\partial x}$ is $O(x^{-\gamma})$ and $o(x^{-\gamma+\epsilon})$ for any $\epsilon>0$. Similarly, $\sum_{i=1}^{n}\frac{\partial f^{\omega}(t,x)}{\partial y_i^{\omega}}\frac{\partial^2 y_i^{\omega}(t,x)}{\partial x^2}$ is $O(x^{-\gamma-1})$ and $o(x^{-\gamma-1+\epsilon})$.

Note that in either case the orders of x are independent of n.

See Appendix A.5 for the proofs. We have seen that optimal portfolio selection is a key question in life and pension insurance, and that numerical implementations of such schemes may approximate the utility function to have sensible first-order conditions. The above result dampens the expectation that such an approximation scheme can lead to "accurate" investment strategies. Indeed, the result points towards further research to understand how significant the deviation from the optimal portfolio is and what the long-term implications for the policyholders are.

4. Conclusion

This paper considers three aspects of expected utility optimisation for the power utility function when approximated by Taylor polynomials. We use methods from asymptotic analysis to understand the effect of the Taylor approximation. This is relevant not only in financial economics but more so in actuarial science as optimising expected utility is a key tool in life and pension insurance, and the approach using Taylor approximations is common in practice.

Working in a Merton-market, the main idea is to decompose the expected utility as

$$\mathbb{E}_{t,x}\left[u(X_T^{\omega})\right] = \sum_{i=0}^{n} \frac{u^{(i)}\left(\mathbb{E}_{t,x}[X_T^{\omega}]\right)}{i!}\mathbb{E}_{t,x}\left[\left(X_T^{\omega} - \mathbb{E}_{t,x}[X_T^{\omega}]\right)^i\right] + R_n^{\omega}(t,x),$$

with explicit remainder term $R_n^{\omega}(t,x)$, and then to investigate the asymptotic properties of $R_n^{\omega}(t,x)$ in x. Here, ω denotes the chosen investment strategy, x is the current wealth of the investor and t denotes time.

Our key results, derived by classical methods from second-order parabolic PDE theory, are threefold:

(i) The asymptotics of $R_n^{\omega}(t,x)$ for large x are independent of the order n of Taylor polynomials and only depend on the parameter γ of the utility function. Indeed, we show that the remainder term is decreasing in x when $\gamma > 1$, and is increasing when $0 < \gamma < 1$. This holds for any admissible strategy ω and not only for the optimal strategy. This gives actuaries a deeper understanding of the quality of the approximation and they can consider this when implementing the method in practice.

(ii) We show that there is an optimal order n that minimises the remainder term for given x and t. This analytically confirms numerical case studies from the literature and also implies that actuaries need to experiment with the order of the Taylor expansion when implementing such optimisation schemes in real-world applications.

(iii) Finally, we consider the approximate investment strategy, i.e. the investment strategy obtained for a truncated Taylor-approximation of the utility function, and compare it to the "true"optimal strategy. As in (i), the quality of the approximation does not increase with n but rather depends on the parameter γ. Again, actuaries need to take this into account if making investment decisions based on such approximations.

The results clearly illustrate the trade-off between feasibility of numerical implementations (e.g. through approximating the utility function) and making "correct" portfolio choices for long-term investments. This is a significant issue for life and pension insurers and indeed for the policyholders whose retirement benefits depend on such portfolio choices. Further research in more general settings both for the market model and the utility function is required to make this more precise. However, our findings clearly indicate issues that actuaries need to be aware of these issues and understand them in more depth.

Appendix

The following sections collect proofs of the main results.

A.1. Properties of certain fundamental solutions

The proofs of our results rely on properties of the probability density of X_t. These are obtained as fundamental solutions of certain partial differential equations (PDEs).

Recall that the Feynman–Kac formula (for example Theorem 7.6, Karatzas and Shreve (1998)) provides the link between the stochastic differential equation (3) and a PDE. Indeed, it guarantees the existence of a function $v(t,x) \in C^{1,2}([0,T]\times\mathbb{R}_+)$ that satisfies the following PDE which is also known as the *Kolmogorov backward equation*:

$$\begin{aligned} -\frac{\partial v(t,x)}{\partial t} &= \frac{1}{2}\omega(t,x)^2\sigma^2x^2\frac{\partial^2 v(t,x)}{\partial x^2} + \Big(r+\omega(t,x)(\mu-r)\Big) \\ &\quad\times x\frac{\partial v(t,x)}{\partial x} \quad \text{and} \\ v(T,x) &= f(x), \end{aligned} \tag{A.1}$$

where $f \in C(\mathbb{R}_+)$ is a given terminal condition.

We need a finer analysis of such solutions and therefore recall the definition of a fundamental solution of such a PDE.

Definition A.1. A *fundamental solution* of a parabolic equation of the form

$$-\frac{\partial v(t,x)}{\partial t} + k(t,x)\,v(t,x) = \frac{1}{2}a^2(t,x)\frac{\partial^2 v(t,x)}{\partial x^2} + b(t,x)\frac{\partial v(t,x)}{\partial x} \tag{A.2}$$

with terminal condition $v(s,x) = f(x)$, where $t \le s$ and $s > 0$, is a non-negative function $\psi(s,y;t,x)$ defined for all $(t,x) \in [0,T]\times\mathbb{R}$ and $(s,y) \in [0,T]\times\mathbb{R}$ and $0 \le t < s \le T$, with following properties:

(i) for fixed $(s,y) \in (0,T]\times\mathbb{R}$, the function ψ viewed as a function of (t,x) satisfies (A.2), and

(ii) for a given terminal condition $f \in C(\mathbb{R})$ and $s \in (0,T]$, we have

$$\lim_{t\to s}\int_{\mathbb{R}}\psi(s,y;t,x)f(y)dy = f(x).$$

We call t the *forward variable* and s the *backward variable*.

The main assertion of this section concerns the existence, uniqueness and bounds of a fundamental solution $\psi(T,y;t,x)$ of (A.1). The bounds are the key to the analysis in this paper.

Proposition A.1. *Under Assumptions 3.1 and 3.2, the following assertions hold:*

(i) For $(s,y),(t,x) \in [0,T)\times\mathbb{R}_+$ with $t < s$, there is a non-negative function $\psi(s,y;t,x)$ that is the unique fundamental solution of (A.1).

(ii) For fixed $T > 0$ the fundamental solution $\psi(T,y;t,x)$ as a function of the backward variables (t,x) satisfies a two-sided bound for the form

$$\frac{1}{Cy\sqrt{T-t}}\exp\left(-\frac{|\ln\frac{y}{x}|^2}{c(T-t)}\right) \le \left|\psi(T,y;t,x)\right| \le \frac{C}{y\sqrt{T-t}}\exp\left(-\frac{c|\ln\frac{y}{x}|^2}{T-t}\right)$$

for some positive constants C and c.

(iii) *There are upper bounds for the derivatives of $\psi(T,y;t,x)$ of the form*

$$\left|\frac{\partial\psi(T,y;t,x)}{\partial x}\right| \le \frac{C}{xy(T-t)}\exp\left(-\frac{c|\ln\frac{y}{x}|^2}{T-t}\right) \quad \text{and}$$
$$\left|\frac{\partial^2\psi(T,y;t,x)}{\partial x^2}\right| \le \frac{C}{x^2 y}\left(\frac{1}{(T-t)^{3/2}}+\frac{1}{T-t}\right) \times \exp\left(-\frac{c|\ln\frac{y}{x}|^2}{T-t}\right),$$

where the constants C and c are as in (ii).

Standard theorems on the existence and uniqueness of fundamental solutions typically require bounded coefficients. In (A.1), both the drift $\big(r+\omega(t,x)(\mu-r)\big)x$ and the volatility $\frac{1}{2}\omega(t,x)^2\sigma^2 x^2$ are unbounded. To achieve bounded coefficients, we make the usual change of variables to log prices: setting $x=e^{\xi}$ with $\xi\in\mathbb{R}$, the domain $\mathbb{R}_+$ is transformed into $\mathbb{R}$. To make this change of variables very transparent, we define two functions of $(t,\xi)\in\mathbb{S}=[0,T]\times\mathbb{R}$ by $\tilde{v}(t,\xi)=v(t,x)$ and $\tilde{\omega}(t,\xi)=\omega(t,x)$.

Proof. Under the change of variables, the parabolic equation (A.1) then becomes

$$-\frac{\partial\tilde{v}(t,\xi)}{\partial t} = \frac{1}{2}\tilde{\omega}(t,\xi)^2\sigma^2\frac{\partial^2\tilde{v}(t,\xi)}{\partial\xi^2} + \left(-\frac{1}{2}\tilde{\omega}(t,\xi)^2\sigma^2+r+\tilde{\omega}(t,\xi)(\mu-r)\right)\frac{\partial\tilde{v}(t,\xi)}{\partial\xi}. \tag{A.3}$$

By Assumption 3.2, we have that $\tilde{\omega}(t,\xi)$ only takes values in the compact interval $[\lambda_1,\lambda_2]$ which is *bounded* away from zero, and $\tilde{\omega}(t,\xi)$ is *Hölder continuous*. Also, the new drift and volatility terms in (A.3) are *bounded* and *Hölder continuous*. Moreover, the volatility term satisfies $\frac{1}{2}\tilde{\omega}(t,\xi)^2\sigma^2 > 0$ hence it is *uniformly parabolic* in $\mathbb{S}$. By Theorem 10, Chapter 1 and Theorem 7, Chapter 9 in Friedman (1964) and Theorem 19 in Porper and Eidelman (1993), we have the following bounds for the fundamental solution of (A.3) and its derivatives.

Under Assumptions 3.1 and 3.2, there is a fundamental solution $\tilde{\psi}(s,\eta;t,\xi)$ of (A.3) with $0\le t<s\le T$ and $\eta,\xi\in\mathbb{R}$. Moreover, the following assertions hold:

(i) The fundamental solution $\tilde{\psi}(s,\eta;t,\xi)$ satisfies a two-sided bound of the form

$$\frac{1}{C\sqrt{s-t}}\exp\left(-\frac{|\eta-\xi|^2}{c(s-t)}\right) \le \left|\tilde{\psi}(s,\eta;t,\xi)\right| \le \frac{C}{\sqrt{s-t}}\exp\left(-\frac{c|\eta-\xi|^2}{s-t}\right)$$

with some positive constants C and c.

(ii) There are upper bounds for the first and second partial derivatives in ξ of $\tilde{\psi}(s,\eta;t,\xi)$ of the form

$$\left|\frac{\partial\tilde{\psi}(s,\eta;t,\xi)}{\partial\xi}\right| \le \frac{C}{s-t}\exp\left(-\frac{c|\eta-\xi|^2}{s-t}\right) \quad \text{and}$$
$$\left|\frac{\partial^2\tilde{\psi}(s,\eta;t,\xi)}{\partial\xi^2}\right| \le \frac{C}{(s-t)^{3/2}}\exp\left(-\frac{c|\eta-\xi|^2}{s-t}\right)$$

with the same constants C and c.

The existence of the fundamental solution $\psi(s,y;t,x)$ of (A.1) follows from simply applying the reverse change of variables $\xi=\ln x$ and $\eta=\ln y$.

The relationship between $\tilde{\psi}(s,\eta;t,\xi)$ and $\psi(s,y;t,x)$ is as follows. For backward variables $(t,\xi)\in[0,T)\times\mathbb{R}$ and $(t,x)\in[0,T)\times\mathbb{R}_+$, the two solutions $\tilde{\psi}(s,\eta;t,\xi)$ and $\psi(s,y;t,x)$ satisfy the relation

$$\tilde{\psi}(s,\eta;t,\xi)=y\cdot\psi(s,y;t,x),$$

which follows from the change of variables.

Finally, this implies the bounds for $\psi(s,y;t,x)$ and its derivatives. Indeed, the two-sided bound for $\psi(s,y;t,x)$ reads

$$\frac{1}{Cy\sqrt{T-t}}\exp\left(-\frac{|\ln y-\ln x|^2}{c(T-t)}\right) \le \left|\psi(T,y;t,x)\right| \le \frac{C}{y\sqrt{T-t}}\exp\left(-\frac{c|\ln\frac{y}{x}|^2}{T-t}\right).$$

And the bounds for its derivatives read

$$\left|\frac{\partial\psi(T,y;t,x)}{\partial x}\right| = x^{-1}y^{-1}\left|\frac{\partial\tilde{\psi}(T,\eta;t,\xi)}{\partial\xi}\right| \le x^{-1}y^{-1}\frac{C}{T-t}\exp\left(-\frac{c|\ln y-\ln x|^2}{T-t}\right) \quad \text{and}$$
$$\left|\frac{\partial^2\psi(T,y;t,x)}{\partial x^2}\right| = x^{-2}y^{-1}\left|\frac{\partial^2\tilde{\psi}(T,\eta;t,\xi)}{\partial\xi^2}-\frac{\partial\tilde{\psi}(T,\eta;t,\xi)}{\partial\xi}\right| \le x^{-2}y^{-1}C\left(\frac{1}{(T-t)^{3/2}}+\frac{1}{T-t}\right) \times \exp\left(-\frac{c|\ln y-\ln x|^2}{T-t}\right).$$

This finishes the proof. □

A.2. Proof of Theorem 3.1

In this part we prove the asymptotics of the remainder term $R_n^{\omega}(t,x)$. We first introduce a proposition that gives bounds for $y_i^{\omega}(t,x)=\mathbb{E}_{t,x}\left[\left(X_T^{\omega}\right)^i\right]$ and their derivatives in x. Note that the non-indexed π represents the circular constant. We keep omitting the superscript ω in this part to simplify the presentation.

Proposition A.2. *Under Assumptions 3.1 and 3.2, the following holds:*

(i) *The functions $y_i(t,x)$ with $1\le i\le n$ have two-sided bounds of the form*

$$x^i \le y_i(t,x) \le C\sqrt{\frac{\pi}{c}}\exp\left\{\frac{1}{4c}i^2(T-t)\right\}\cdot x^i$$

with some positive constants C and c the same as in Proposition A.1.

(ii) *There are upper bounds for their derivatives of the form*

$$\left|\frac{\partial y_i(t,x)}{\partial x}\right| \le C\sqrt{\frac{\pi}{c}}\frac{1}{\sqrt{T-t}}\exp\left\{\frac{1}{4c}i^2(T-t)\right\}\cdot x^{i-1} \quad \text{and}$$
$$\left|\frac{\partial^2 y_i(t,x)}{\partial x^2}\right| \le C\sqrt{\frac{\pi}{c}}\left(\frac{1}{T-t}+\frac{1}{\sqrt{T-t}}\right) \times \exp\left\{\frac{1}{4c}i^2(T-t)\right\}\cdot x^{i-2}$$

with the same constants C and c in (i).

The lower bounds in (i) are implied by the minimum principle for $y_i(t,x)$. Intuitively, the minimum principle implies that the wealth x of an investor at current time t is expected to grow with time. While we prove the lemma in the PDE-setting, it also follows from a sub-martingale argument.

Proof. We start with the lower bounds where without loss of generality $i = 1$. Note that $y_1(t,x) = \varphi(t,x)$. By Theorem 10, Chapter 2 in Friedman (1964), the function $\varphi(t,x)$ is a solution of (A.1) subject to the terminal condition $\varphi(T,x) = x$. Define an operator L acting on functions $v \in C^{1,2}([0,T]\times\mathbb{R})$ by

$$L(v) = a^2(t,x)\frac{\partial^2 v}{\partial x^2} + b(t,x)\frac{\partial v}{\partial x} + \frac{\partial v}{\partial t},$$

where $a(t,x) > 0$ and $b(t,x) > 0$ are the coefficients in (A.1). Clearly, we have $L(\varphi) = 0$. Choose a function $v(t,x) = \varphi(t,x) - x$ with terminal condition $v(T,x) = \varphi(T,x) - x = 0$. Then we have for $(t,x) \in [0,T]\times\mathbb{R}_+$

$$L(v(t,x)) = L(\varphi(t,x) - x) = L(\varphi(t,x)) - b(t,x) < 0.$$

We extend the domain from $[0,T]\times\mathbb{R}_+$ to $[0,T]\times\mathbb{R}_+\cup\{0\}$ by giving a boundary condition that $v(t,0) = 0$. By Proposition A.2, the growth of $v(t,x)$ is bounded because the growth of $\varphi(t,x)$ is bounded by

$$\varphi(t,x) \le C\sqrt{\frac{\pi}{c}}e^{\frac{T-t}{4c}}\cdot x.$$

Then by Theorem 1.1 in Chen et al. (1973), $L(v) < 0$ implies $v(t,x) \ge 0$ in $[0,T]\times\mathbb{R}_+$. Hence we have $\varphi(t,x) \ge x$.

We now prove the upper bounds. Recall that the $y_i(t,x)$ satisfy (A.1). By Theorem 10 of Chapter 2 in Friedman (1964), Proposition A.1(ii) and standard properties of Gaussian integrals, we have that

$$\begin{aligned} y_i(t,x) &= \int_0^\infty y^i\cdot\psi(T,y;t,x)dy \\ &\le \int_0^\infty y^i\cdot y^{-1}\frac{C}{\sqrt{T-t}}e^{-\frac{c}{T-t}|\ln y-\ln x|^2}dy \\ &= \frac{C}{\sqrt{T-t}}\int_{-\infty}^\infty e^{i\eta}\cdot e^{-\frac{c}{T-t}|\eta-\xi|^2}d\eta \\ &= C\sqrt{\frac{\pi}{c}}e^{\frac{i^2(T-t)}{4c}}\cdot x^i. \end{aligned}$$

By Proposition A.1(iii), the derivatives of $y_i(t,x)$ satisfy

$$\begin{aligned} \left|\frac{\partial y_i(t,x)}{\partial x}\right| &\le \int_0^\infty y^i\left|\frac{\partial\psi(T,y;t,x)}{\partial x}\right|dy \\ &\le \int_0^\infty y^i x^{-1}y^{-1}\frac{C}{T-t}\exp\left(-\frac{c|\ln\frac{y}{x}|^2}{T-t}\right)dy \\ &= C\sqrt{\frac{\pi}{c}}\frac{1}{\sqrt{T-t}}e^{\frac{i^2(T-t)}{4c}}x^{i-1}, \\ \left|\frac{\partial^2 y_i(t,x)}{\partial x^2}\right| &\le \int_0^\infty y^i x^{-2}y^{-1}C\left(\frac{1}{(T-t)^{3/2}}+\frac{1}{(T-t)}\right) \\ &\quad\times\exp\left(-\frac{c|\ln\frac{y}{x}|^2}{T-t}\right)dy \\ &= C\sqrt{\frac{\pi}{c}}\left(\frac{1}{T-t}+\frac{1}{\sqrt{T-t}}\right)e^{\frac{i^2(T-t)}{4c}}x^{i-2}. \end{aligned}$$

This proves the claim. □

The next lemma contains two useful inequalities.

Lemma A.1. *For any $0 \le z \le 1$, $k \ge 1$, $x \ge 0$ and $y \ge 0$ we have that $|(y-x)^k| \le y^k + x^k$ and $|(zy+(1-z)x)^{-k}| \le y^{-k} + x^{-k}$.*

Proof. If $y > x$, then $\left|(y-x)^k\right| = (y-x)^k \le y^k \le y^k + x^k$. If $y \le x$, then $\left|(y-x)^k\right| = (x-y)^k \le x^k + y^k$, yielding the first inequality. For the second note that $zy + (1-z)x$ takes values between x and y. By varying $z \in [0,1]$, the maximum of the expression is $\max\{x^{-k}, y^{-k}\} \le y^{-k} + x^{-k}$. □

We are now ready to prove Theorem 3.1.

Proof. Let $\psi(T,y;t,x)$ be the fundamental solution of (A.1) and we obtain

$$\begin{aligned} R_n(t,x) = \binom{-\gamma}{n}\int_0^1(1-z)^n\int_0^\infty &\frac{(y-\varphi(t,x))^{n+1}}{(zy+(1-z)\varphi(t,x))^{\gamma+n}} \\ &\times\psi(T,y;t,x)dydz. \end{aligned} \tag{A.4}$$

We first estimate the double integral in (A.4). By Propositions A.1 (ii) and A.2, Lemma A.1 and standard results on Gaussian integrals, we find

$$\begin{aligned} &\left|\int_0^\infty\big(y-\varphi(t,x)\big)^{n+1}\big(zy+(1-z)\varphi(t,x)\big)^{-\gamma-n}\psi(T,y;t,x)dy\right| \\ &\le \int_0^\infty\left(y^{1-\gamma}+y^{n+1}\varphi(t,x)^{-\gamma-n}+y^{-\gamma-n}\varphi(t,x)^{n+1}+\varphi(t,x)^{1-\gamma}\right) \\ &\quad\times\ \psi(T,y;t,x)dy \\ &\le \int_0^\infty y^{1-\gamma}\psi(T,y;t,x)dy + \varphi(t,x)^{-\gamma-n}\int_0^\infty y^{n+1}\psi(T,y;t,x)dy \\ &\quad+\varphi(t,x)^{n+1}\int_0^\infty y^{-\gamma-n}\psi(T,y;t,x)dy+\varphi(t,x)^{1-\gamma} \\ &\le C\sqrt{\frac{\pi}{c}}e^{(1-\gamma)^2\frac{T-t}{4c}}x^{1-\gamma}+\varphi(t,x)^{-\gamma-n}C\sqrt{\frac{\pi}{c}}e^{(n+1)^2\frac{T-t}{4c}}x^{n+1} \\ &\quad+\varphi(t,x)^{n+1}C\sqrt{\frac{\pi}{c}}e^{(-\gamma-n)^2\frac{T-t}{4c}}x^{-\gamma-n}+\varphi(t,x)^{1-\gamma} \\ &\le \Bigg(C\sqrt{\frac{\pi}{c}}e^{(1-\gamma)^2\frac{T-t}{4c}}+C\sqrt{\frac{\pi}{c}}e^{(n+1)^2\frac{T-t}{4c}}\left(\frac{\varphi(t,x)}{x}\right)^{-\gamma-n} \\ &\quad+C\sqrt{\frac{\pi}{c}}e^{(-\gamma-n)^2\frac{T-t}{4c}}\left(\frac{\varphi(t,x)}{x}\right)^{n+1}+\left(\frac{\varphi(t,x)}{x}\right)^{1-\gamma}\Bigg)\cdot x^{1-\gamma}. \end{aligned}$$

Note that the term $\frac{\varphi(t,x)}{x}$ has a two sided bound of the form $1 \le \frac{\varphi(t,x)}{x} \le C\sqrt{\frac{\pi}{c}}e^{\frac{1}{4c}(T-t)}$. Then there is a function $M_1(t,n,\gamma)$ independent of x that satisfies

$$\begin{aligned} &\left|\int_0^1(1-z)^n\int_0^\infty\big(y-\varphi(t,x)\big)^{n+1}\big(zy+(1-z)\varphi(t,x)\big)^{-\gamma-n}\right. \\ &\quad\left.\times\ \psi(T,y;t,x)dydz\right| \\ &\le \int_0^1(1-z)^n dz\Bigg(C\sqrt{\frac{\pi}{c}}e^{(1-\gamma)^2\frac{T-t}{4c}}+C\sqrt{\frac{\pi}{c}}e^{(n+1)^2\frac{T-t}{4c}} \\ &\quad\times\left(\frac{\varphi(t,x)}{x}\right)^{-\gamma-n} \\ &\quad+C\sqrt{\frac{\pi}{c}}e^{(-\gamma-n)^2\frac{T-t}{4c}}\left(\frac{\varphi(t,x)}{x}\right)^{n+1}+\left(\frac{\varphi(t,x)}{x}\right)^{1-\gamma}\Bigg)\cdot x^{1-\gamma} \\ &= M_1(t,n,\gamma)x^{1-\gamma}. \end{aligned} \tag{A.5}$$

Here the order of x is independent of n.

To prove $R_n(t,x) \sim M(t)\cdot\varphi(t,x)^{1-\gamma}$, it suffices to show that for fixed t the limit

$$\lim_{x\to\infty}\frac{R_n(t,x)}{\varphi(t,x)^{1-\gamma}} = M(t)$$

exists and is independent of x. We extract $\varphi(t,x)^{1-\gamma}$ from the double integral of $R_n(t,x)$ and it becomes

$$R_n(t,x) = \binom{-\gamma}{n}\varphi(t,x)^{1-\gamma}$$

$$\times\int_0^1 (1-z)^n \int_0^\infty \left(\frac{y}{\varphi(t,x)}-1\right)^{n+1}\left(z\frac{y}{\varphi(t,x)}+(1-z)\right)^{-\gamma-n}$$
$$\times\ \psi(T,y;t,x)dydz.$$

Now apply a change of variables $\hat{y}=\frac{y}{\varphi(t,x)}$. The double integral becomes

$$\int_0^1 (1-z)^n \int_0^\infty (\hat{y}-1)^{n+1}\left(z\hat{y}+(1-z)\right)^{-\gamma-n}\psi(T,\hat{y}\varphi(t,x);t,x)$$
$$\times\varphi(t,x)d\hat{y}dz.$$

By defining $\hat{\psi}(T,\hat{y};t,x):=\psi(T,\hat{y}\varphi(t,x);t,x)\varphi(t,x)$ this is

$$\int_0^1 (1-z)^n \int_0^\infty (\hat{y}-1)^{n+1}\left(z\hat{y}+(1-z)\right)^{-\gamma-n}\hat{\psi}(T,\hat{y};t,x)d\hat{y}dz. \tag{A.6}$$

Proposition A.1(ii) implies an upper bound for $\hat{\psi}(T,\hat{y};t,x)$ of the form

$$\begin{aligned}\hat{\psi}(T,\hat{y};t,x) &\le \frac{C\varphi(t,x)}{y\sqrt{T-t}}\exp\left(-\frac{c|\ln\frac{y}{x}|^2}{T-t}\right)\\ &= \frac{C}{\hat{y}\sqrt{T-t}}\exp\left(-\frac{c|\ln\hat{y}\frac{\varphi(t,x)}{x}|^2}{T-t}\right)\\ &\le \frac{C}{\hat{y}\sqrt{T-t}}\exp\left(-\frac{c|\ln\hat{y}|^2}{T-t}\right)\end{aligned}$$

for constants C and c. Similarly to (A.5), one can derive a bound for (A.6) that reads

$$\left|\int_0^1 (1-z)^n \int_0^\infty (\hat{y}-1)^{n+1}\left(z\hat{y}+(1-z)\right)^{-\gamma-n}\hat{\psi}(T,\hat{y};t,x)d\hat{y}dz\right|$$
$$\le\hat{M}_1(t,n,\gamma),$$

which is independent of x so that in particular the double integral exists for any $x\in\mathbb{R}_+$. Note that $\hat{y}\to 0$ when $x\to\infty$ and the bound for $\hat{\psi}(T,\hat{y};t,x)$ goes to zero because the exponential part decays faster than $\hat{y}^{-1}$. Then the limit

$$h(T,\hat{y};t)=\lim_{x\to\infty}\hat{\psi}(T,\hat{y};t,x)$$

exists and is bounded. By Lebesgue's Dominated Convergence Theorem, the limit of the double integral satisfies

$$\begin{aligned}&\lim_{x\to\infty}\int_0^1 (1-z)^n \int_0^\infty (\hat{y}-1)^{n+1}\left(z\hat{y}+(1-z)\right)^{-\gamma-n}\\ &\quad\times\ \hat{\psi}(T,\hat{y};t,x)d\hat{y}dz\\ &=\int_0^1 (1-z)^n \int_0^\infty (\hat{y}-1)^{n+1}\left(z\hat{y}+(1-z)\right)^{-\gamma-n}\\ &\quad\times\ \lim_{x\to\infty}\hat{\psi}(T,\hat{y};t,x)d\hat{y}dz\\ &=\int_0^1 (1-z)^n \int_0^\infty (\hat{y}-1)^{n+1}\left(z\hat{y}+(1-z)\right)^{-\gamma-n}h(T,\hat{y};t)d\hat{y}dz,\end{aligned}$$

which is independent of x. We define $M(t)\in C^1([0,T])$ as a function that reads

$$M(t)=\binom{-\gamma}{n}\int_0^1 (1-z)^n \int_0^\infty (\hat{y}-1)^{n+1}\left(z\hat{y}+(1-z)\right)^{-\gamma-n}$$
$$\times h(T,\hat{y};t)d\hat{y}dz,$$

which is also independent of x. Then we have $\lim_{x\to\infty}\frac{R_n(t,x)}{\varphi(t,x)^{1-\gamma}}=M(t)$. Hence $R_n(t,x)$ satisfies $R_n(t,x)\sim M(t)\cdot\varphi(t,x)^{1-\gamma}$.

Note that as we omit the superscript ω in this proof, we emphasise that the function $M(t)$ depends on ω as well. This dependence comes from the limit of the fundamental solution and $\hat{y}$.

To prove $R_n(t,x)=O(x^{1-\gamma})$ note that from (A.5) we have

$$\left|\int_0^1 (1-z)^n \int_0^\infty \left(y-\varphi(t,x)\right)^{n+1}\left(zy+(1-z)\varphi(t,x)\right)^{-\gamma-n}\right.$$
$$\left.\times\ \psi(T,y;t,x)dydz\right|$$
$$\le M_1(t,n,\gamma)x^{1-\gamma},$$

where $M_1(t)$ is independent of x. Insert it into $R_n(t,x)$ and we have that

$$|R_n(t,x)|\le\left|\frac{1}{1-\gamma}\binom{-\gamma}{n}\right|M_1(t,n,\gamma)x^{1-\gamma}.$$

Hence the term $\frac{R_n(t,x)}{x^{1-\gamma}}$ is bounded by $\left|\frac{R_n(t,x)}{x^{1-\gamma}}\right|\le\left|\frac{1}{1-\gamma}\binom{-\gamma}{n}\right|M_1(t,n,\gamma)$, where the bounds are independent of x. Moreover, for any $\epsilon>0$, we have the estimate

$$\lim_{x\to\infty}\left|\frac{R_n(t,x)}{x^{1-\gamma+\epsilon}}\right|\le\left|\frac{1}{1-\gamma}\binom{-\gamma}{n}\right|M_1(t,n,\gamma)\cdot\lim_{x\to\infty}x^{-\epsilon}=0.$$

Hence $\frac{R_n(t,x)}{x^{1-\gamma+\epsilon}}\to 0$ for any $\epsilon>0$ as $x\to\infty$, which is independent of n. $\square$

A.3. Proof of Theorem 3.2

Proof. We keep omitting the superscript ω in this part to simplify the presentation. Note that the double integral in $R_n(t,x)$ from (A.4) is positive, and the sign of $R_n(t,x)$ depends on the sign of $\binom{-\gamma}{n}$ i.e. on the parity of n: $R_n(t,x)$ is negative when n is odd and positive when n is even. This suggests to distinguish two cases according to the parity of n. We only consider the case n odd as n even is similar.

So assume that n odd. Here we rewrite the remainder term as

$$\begin{aligned}&R_n(t,x)\\ &=\frac{\mathbb{E}_{t,x}\left[X_T^{1-\gamma}\right]}{1-\gamma}-\frac{\varphi^{1-\gamma}}{1-\gamma}\left(1+\binom{1-\gamma}{2}\frac{m_2}{\varphi^2}+\cdots\right.\\ &\quad\left.+\binom{1-\gamma}{n}\frac{m_n}{\varphi^n}\right)\\ &=\frac{\mathbb{E}_{t,x}\left[X_T^{1-\gamma}\right]}{1-\gamma}-\varphi^{1-\gamma}\left(1+(-1)\frac{\gamma}{2}\frac{m_2}{\varphi^2}+\cdots\right.\\ &\quad\left.+(-1)^{n-1}\frac{1}{n}\binom{\gamma+n-2}{n-1}\frac{m_n}{\varphi^n}\right),\end{aligned}$$

where m_i are the ith central moment of X_T and φ is the mean of X_T. The increment of the absolute values of the remainder terms reads

$$\begin{aligned}&|R_{n+2}(t,x)|-|R_n(t,x)|\\ &=R_n(t,x)-R_{n+2}(t,x)\\ &=\varphi^{1-\gamma}\left((-1)^n\frac{1}{n+1}\binom{\gamma+n-1}{n}\frac{m_{n+1}}{\varphi^{n+1}}\right.\\ &\quad\left.+(-1)^{n+1}\frac{1}{n+2}\binom{\gamma+n}{n+1}\frac{m_{n+2}}{\varphi^{n+2}}\right)\\ &=\varphi^{1-\gamma}\frac{1}{n+1}\binom{\gamma+n-1}{n}\left(-\frac{m_{n+1}}{\varphi^{n+1}}+\frac{\gamma+n}{n+2}\frac{m_{n+2}}{\varphi^{n+2}}\right).\end{aligned}$$

Then we have that $|R_{n+2}(t,x)|-|R_n(t,x)|>0$ if and only if

$$\frac{m_{n+2}/\varphi^{n+2}}{m_{n+1}/\varphi^{n+1}}>\frac{n+2}{\gamma+n}. \tag{A.7}$$

In the notation introduced immediately before (A.6), this can be written as

$$\frac{m_n}{\varphi^n} = \int_0^\infty (\hat{y}-1)^n \psi(\varphi\hat{y})\varphi d\hat{y} = \int_0^\infty (\hat{y}-1)^n \hat{\psi}(\hat{y}) d\hat{y}.$$

We split the integral $\int_0^\infty (\hat{y}-1)^n \hat{\psi}(\hat{y})d\hat{y}$ into three parts:

(i) The first part is $\int_0^1 (\hat{y}-1)^n \hat{\psi}(\hat{y})d\hat{y}$. As n is odd, this integral is negative as it is equal to $-\int_0^1 (1-\hat{y})^n \hat{\psi}(\hat{y})d\hat{y}$. The integral tends to zero as $n \to \infty$.
(ii) The second part is $\int_1^2 (\hat{y}-1)^n \hat{\psi}(\hat{y})d\hat{y}$, which is positive and also tends to zero as n becomes larger.
(iii) The third part is $\int_2^\infty (\hat{y}-1)^n \hat{\psi}(\hat{y})d\hat{y}$. This integral is positive and growing as $n \to \infty$.

In summary, the integral $\int_0^\infty (\hat{y}-1)^n \hat{\psi}(\hat{y})d\hat{y}$ grows in n as $n \to \infty$. Hence $\frac{m_{n+2}/\varphi^{n+2}}{m_{n+1}/\varphi^{n+1}}$ increases in n as $n \to \infty$, and from (A.7) we deduce that the absolute value of the remainder term increases for n sufficiently large. □

A.4. Proof of Theorem 3.3

In this part we follow the ideas and steps of the proof of Theorem 1 in Kryger et al. (2019) to give the proof of Theorem 3.3.

Proof. For our optimisation problem, we define a value function $J^\omega(t,x)$ of any admissible strategy ω that reads

$$J^\omega(t,x) = f\left(t,x,y_1^\omega(t,x),\ldots,y_n^\omega(t,x)\right) + R_n^\omega(t,x).$$

Recall that the expected remainder term $R_n^\omega(t,x)$ reads

$$\begin{aligned} &R_n^\omega(t,x) \\ &= \int_0^\infty \left(y-\varphi^\omega(t,x)\right)^{n+1} u^{(n+1)}\left(z\left(y-\varphi^\omega(t,x)\right)+\varphi^\omega(t,x)\right) \\ &\quad \times \psi^\omega(T,y;t,x)\,dy. \end{aligned} \tag{A.8}$$

By chain rules we have that

$$\frac{\partial J^\omega(t,x)}{\partial t} = \frac{\partial f^\omega(t,x)}{\partial t} + \sum_{i=1}^n \frac{\partial f^\omega(t,x)}{\partial y_i}\frac{\partial y_i^\omega(t,x)}{\partial t} + \frac{\partial R_n^\omega(t,x)}{\partial t}.$$

We derive the term $\frac{\partial R_n^\omega(t,x)}{\partial t}$ from (A.8) and it reads

$$\begin{aligned} &\frac{\partial R_n^\omega(t,x)}{\partial t} \\ &= \frac{1}{n!}\int_0^1 (1-z)^n \\ &\left\{\frac{\partial\varphi^\omega(t,x)}{\partial t}\int_0^\infty \left[(-n-1)\left(y-\varphi^\omega(t,x)\right)^n\right.\right. \\ &\quad \times u^{(n+1)}\left(zy+(1-z)\varphi^\omega(t,x)\right) \\ &\quad \left. +(1-z)\left(y-\varphi^\omega(t,x)\right)^{n+1} u^{(n+2)}\left(zy+(1-z)\varphi^\omega(t,x)\right)\right] \\ &\quad \times \psi^\omega(T,y;t,x)dy \\ &\quad + \int_0^\infty \left(y-\varphi^\omega(t,x)\right)^{n+1} u^{(n+1)}\left(zy+(1-z)\varphi^\omega(t,x)\right) \\ &\quad \left.\times \frac{\partial\psi^\omega(T,y;t,x)}{\partial t}dy\right\}dz. \end{aligned}$$

Recall that the dynamics of X_t^ω reads

$$dX_t^\omega = \left[r+\omega(t,x)(\mu-r)\right]X_t^\omega dt + \omega(t,x)\sigma X_t^\omega dW_t.$$

By Theorem 7.6 in Karatzas and Shreve (1998), the functions $\varphi^\omega(t,x)$ and $\psi^\omega(T,y;t,x)$ satisfy the PDEs

$$\begin{aligned} -\frac{\partial\varphi^\omega(t,x)}{\partial t} &= \frac{1}{2}\omega(t,x)^2\sigma^2x^2\frac{\partial^2\varphi^\omega(t,x)}{\partial x^2} \\ &\quad + \left(r+\omega(t,x)(\mu-r)\right)x\frac{\partial\varphi^\omega(t,x)}{\partial x}, \\ \varphi^\omega(T,x) &= x \quad \text{and} \\ -\frac{\partial\psi^\omega(T,y;t,x)}{\partial t} &= \frac{1}{2}\omega(t,x)^2\sigma^2x^2\frac{\partial^2\psi^\omega(T,y;t,x)}{\partial x^2} \\ &\quad + \left(r+\omega(t,x)(\mu-r)\right)x\frac{\partial\psi^\omega(T,y;t,x)}{\partial x}. \end{aligned}$$

A routine calculation yields

$$\begin{aligned} \frac{\partial R_n^\omega(t,x)}{\partial t} &= -\left(\frac{1}{2}\omega(t,x)^2\sigma^2x^2R_{2,n}^\omega(t,x) + \left(r+\omega(t,x)(\mu-r)\right)\right. \\ &\quad \left.\times xR_{1,n}^\omega(t,x)\right), \end{aligned}$$

with $R_{1,n}^\omega$ and $R_{2,n}^\omega$ given explicitly by

$$\begin{aligned} &R_{1,n}^\omega(t,x) \\ &= \frac{1}{n!}\int_0^1 (1-z)^n \\ &\left\{\frac{\partial\varphi^\omega(t,x)}{\partial x}\int_0^\infty \left[(-n-1)\left(y-\varphi^\omega(t,x)\right)^n\right.\right. \\ &\quad \times u^{(n+1)}\left(zy+(1-z)\varphi^\omega(t,x)\right) \\ &\quad \left. +(1-z)\left(y-\varphi^\omega(t,x)\right)^{n+1} u^{(n+2)}\left(zy+(1-z)\varphi^\omega(t,x)\right)\right] \\ &\quad \times \psi^\omega(T,y;t,x)dy \\ &\quad + \int_0^\infty \left(y-\varphi^\omega(t,x)\right)^{n+1} u^{(n+1)}\left(zy+(1-z)\varphi^\omega(t,x)\right) \\ &\quad \left.\times \frac{\partial\psi^\omega(T,y;t,x)}{\partial x}dy\right\}dz \end{aligned} \tag{A.9}$$

and

$$\begin{aligned} &R_{2,n}^\omega(t,x) \\ &= \frac{1}{n!}\int_0^1 (1-z)^n \\ &\left\{\frac{\partial^2\varphi^\omega(t,x)}{\partial x^2}\int_0^\infty \left[(-n-1)\left(y-\varphi^\omega(t,x)\right)^n\right.\right. \\ &\quad \times u^{(n+1)}\left(zy+(1-z)\varphi^\omega(t,x)\right) \\ &\quad \left. +(1-z)\left(y-\varphi^\omega(t,x)\right)^{n+1} u^{(n+2)}\left(zy+(1-z)\varphi^\omega(t,x)\right)\right] \\ &\quad \times \psi^\omega(T,y;t,x)dy \\ &\quad + \int_0^\infty \left(y-\varphi^\omega(t,x)\right)^{n+1} u^{(n+1)}\left(zy+(1-z)\varphi^\omega(t,x)\right) \\ &\quad \left.\times \frac{\partial^2\psi^\omega(T,y;t,x)}{\partial x^2}dy\right\}dz. \end{aligned} \tag{A.10}$$

Then the t-derivative of the value function $J^{\omega}(t,x)$ satisfies

$$\frac{\partial J^{\omega}(t,x)}{\partial t} = \frac{\partial f^{\omega}(t,x)}{\partial t} - \frac{1}{2}\omega(t,x)^2\sigma^2 x^2 \left(\sum_{i=1}^{n} \frac{\partial f^{\omega}(t,x)}{\partial y_i}\frac{\partial^2 y_i^{\omega}(t,x)}{\partial x^2} + R_{2,n}^{\omega}(t,x)\right) - \Big(r+\omega(t,x)(\mu-r)\Big)x\left(\sum_{i=1}^{n}\frac{\partial f^{\omega}(t,x)}{\partial y_i}\frac{\partial y_i^{\omega}(t,x)}{\partial x} + R_{1,n}^{\omega}(t,x)\right).$$

Next, we follow the steps of the proof of Theorem 1 in Kryger et al. (2019) with functions $G_i^{\omega^*}(t,x)$ and $H_n^{\omega^*}(t,x)$ that satisfy

$$-\frac{\partial G_i^{\omega^*}(t,x)}{\partial t} = \big(r+\omega^*(t,x)(\mu-r)\big)x\frac{\partial G_i^{\omega^*}(t,x)}{\partial x} + \frac{1}{2}\sigma^2\omega^*(t,x)^2x^2\frac{\partial^2 G_i^{\omega^*}(t,x)}{\partial x^2},$$
$$-\frac{\partial H_n^{\omega^*}(t,x)}{\partial t} = \big(r+\omega^*(t,x)(\mu-r)\big)x\frac{\partial H_n^{\omega^*}(t,x)}{\partial x} + \frac{1}{2}\sigma^2\omega^*(t,x)^2x^2\frac{\partial^2 H_n^{\omega^*}(t,x)}{\partial x^2},$$
$$G_i^{\omega^*}(T,x) = g_i(x) \quad \text{and}$$
$$H_n^{\omega^*}(T,x) = 0.$$

Omitting some steps that are detailed in Kryger et al. (2019), the optimal strategy $\omega^*(t,x)$ is determined by a first-order condition from

$$\omega^*(t,x) = \arg\inf\left\{ -\frac{1}{2}\omega(t,x)^2\sigma^2x^2\left(\sum_{i=1}^{n}\frac{\partial f(t,x)}{\partial y_i}\frac{\partial^2 G_i^{\omega^*}(t,x)}{\partial x^2} + \frac{\partial^2 H_n^{\omega^*}(t,x)}{\partial x^2}\right) - \Big(r+\omega(t,x)(\mu-r)\Big)x\left(\sum_{i=1}^{n}\frac{\partial f(t,x)}{\partial y_i}\frac{\partial G_i^{\omega^*}(t,x)}{\partial x} + \frac{\partial H_n^{\omega^*}(t,x)}{\partial x}\right)\right\},$$

where we assume $\sum_{i=1}^{n}\frac{\partial f(t,x)}{\partial y_i}\frac{\partial^2 G_i^{\omega^*}(t,x)}{\partial x^2} + \frac{\partial^2 H_n^{\omega^*}(t,x)}{\partial x^2} < 0$ to ensure that ω^* is the optimum and the equalities $\frac{\partial H_n^{\omega^*}(t,x)}{\partial x} = R_{1,n}^{\omega^*}(t,x)$ and $\frac{\partial^2 H_n^{\omega^*}(t,x)}{\partial x^2} = R_{2,n}^{\omega^*}(t,x)$ hold only with the optimal strategy ω^*. □

A.5. Proof of Theorem 3.4

In this part we give the asymptotics of $R_{1,n}^{\omega}(t,x)$ and $R_{2,n}^{\omega}(t,x)$. The proofs are similar to the proof of Theorem 3.1. We omit the superscript ω again to simplify the presentation.

Proof. Recall that the two terms $R_{1,n}(t,x)$ and $R_{2,n}(t,x)$ are explicitly given in (A.9) and (A.9). We must estimate the integrals to extract the growth in x. Similarly to (A.5), we have that

$$\left|\int_0^1(1-z)^n\int_0^{\infty}\frac{(y-\varphi(t,x))^{n+1}}{(zy+(1-z)\varphi(t,x))^{\gamma+n}}\frac{\partial\psi(T,y;t,x)}{\partial x}dydz\right| \le M_{1,1}(t,n,\gamma)x^{-\gamma} \tag{A.11}$$

and

$$\left|\int_0^1(1-z)^n\int_0^{\infty}\frac{(y-\varphi(t,x))^{n+1}}{(zy+(1-z)\varphi(t,x))^{\gamma+n}}\frac{\partial^2\psi(T,y;t,x)}{\partial x^2}dydz\right| \le M_{2,1}(t,n,\gamma)x^{-1-\gamma}. \tag{A.12}$$

for some functions $M_{1,1}$ and $M_{2,1}$ that are independent of x. The previously developed arguments yield a function M_0 independent of x such that

$$\begin{aligned}
&\left|R_{1,n}(t,x)\right|\left|\binom{-\gamma}{n}\right|^{-1} \\
&\le \left|\frac{\partial\varphi(t,x)}{\partial x}\right|\left|\int_0^1(1-z)^n\int_0^{\infty}(n+1)\big(y-\varphi(t,x)\big)^n \times \big(zy+(1-z)\varphi(t,x)\big)^{-\gamma-n} + (1-z)(\gamma+n)\big(y-\varphi(t,x)\big)^{n+1}\big(zy+(1-z)\varphi(t,x)\big)^{-\gamma-n-1} \times \psi(T,y;t,x)dydz\right| \\
&\quad + \left|\int_0^1(1-z)^n\int_0^{\infty}\big(y-\varphi(t,x)\big)^{n+1}\big(zy+(1-z)\varphi(t,x)\big)^{-\gamma-n} \times \frac{\partial\psi(T,y;t,x)}{\partial x}dydz\right| \\
&\le C\sqrt{\frac{\pi}{c}}\frac{1}{\sqrt{T-t}}\Big((n+1)M_0(t,n,\gamma)+(\gamma+n)M_0(t,(n+1),\gamma)\Big) \times e^{\frac{T-t}{4c}}x^{-\gamma} + M_{1,1}(t,n,\gamma)x^{-\gamma},
\end{aligned}$$

and likewise with the second derivative $\frac{\partial^2\psi(T,y;t,x)}{\partial x^2}$ one has

$$\begin{aligned}
&\left|R_{2,n}(t,x)\right|\left|\binom{-\gamma}{n}\right|^{-1} \\
&\le C\sqrt{\frac{\pi}{c}}\left(\frac{1}{T-t}+\frac{1}{\sqrt{T-t}}\right)\Big((n+1)M_0(t,n,\gamma) + (\gamma+n)M_0(t,n+1,\gamma)\Big) \times e^{\frac{T-t}{4c}}x^{-\gamma-1} + M_{2,1}(t,n,\gamma)x^{-\gamma-1}.
\end{aligned}$$

Hence we have by combining the terms without x that $\left|R_{1,n}(t,x)\right| \le M_3(t,n,\gamma)\cdot x^{-\gamma}$ and $\left|R_{2,n}(t,x)\right| \le M_4(t,n,\gamma)\cdot x^{-\gamma-1}$. Here M_3 and M_4 are functions independent of x, and the orders of x are independent of n. Then the asymptotics in Theorem 3.4.(i) follow.

Next we prove Theorem 3.4.(ii). For the two summations, recall that f can be expressed as

$$\begin{aligned}
f(t,x,y_1(t,x),\dots,y_n(t,x)) &= \binom{-\gamma}{n}\frac{(-1)^{n+1}}{-\gamma}\frac{1+(1-\gamma)(n-1)}{1-\gamma} \times y_1(t,x)^{1-\gamma} \\
&\quad + \binom{-\gamma}{n}\sum_{i=2}^{n}\binom{n}{i}\frac{(-1)^{n+i}}{1-\gamma-i} \times y_1(t,x)^{1-\gamma-i}y_i(t,x) \\
&\quad + R_n(t,x).
\end{aligned}$$

By Proposition A.2, for $i=1$ we have that

$$\left|\frac{\partial f(t,x)}{\partial y_1}\right| \le \left|\binom{-\gamma}{n}\frac{(-1)^{n+1}}{-\gamma}\big(1+(1-\gamma)(n-1)\big)y_1(t,x)^{-\gamma}\right| + \left|\binom{-\gamma}{n}\sum_{i=2}^{n}\binom{n}{i}(-1)^{n+i}y_1^{\omega}(t,x)^{-\gamma-i}y_i^{\omega}(t,x)\right|$$

$$\leq \left| \binom{-\gamma}{n} \frac{(-1)^{n+1}}{-\gamma} \big(1 + (1-\gamma)(n-1)\big) \right| x^{-\gamma} + \left| \binom{-\gamma}{n} \sum_{i=2}^{n} \binom{n}{i} (-1)^{n+i} \right| C \sqrt{\frac{\pi}{c}} e^{\frac{1}{4c} i^2 (T-t)} x^i x^{-\gamma-i} = M_1^{(y)}(t, n, \gamma) \cdot x^{-\gamma},$$

for a suitable function $M_1^{(y)}$. For $2 \leq i \leq n$ we have analogously that

$$\left| \frac{\partial f(t,x)}{\partial y_i} \right| \leq \left| \binom{-\gamma}{n} \sum_{i=2}^{n} \binom{n}{i} \frac{(-1)^{n+i}}{1-\gamma-i} \right| x^{1-\gamma-i} = M_i^{(y)}(t, n, \gamma) \cdot x^{1-\gamma-i}.$$

The desired asymptotics then follow using the bounds for $\frac{\partial y_i(t,x)}{\partial x}$ from Proposition A.2 and collecting terms. □

References

Bernhardt, T., Donnelly, C., 2019. Modern tontine with bequest: Innovation in pooled annuity products. Insurance Math. Econom. 86, 168–188.

Björk, T., Murgoci, A., 2010. A general theory of markovian time inconsistent stochastic control problems. SSRN Electron. J..

Brandt, M.W., Goyal, A., Santa-Clara, P., Stroud, J.R., 2005. A simulation approach to dynamic portfolio choice with an application to learning about return predictability. Rev. Financ. Stud. 18 (3), 831–873.

Brockett, P.L., Garven, J.R., 1998. A reexamination of the relationship between preferences and moment orderings by rational risk-averse investors. Geneva Pap. Risk Insur. Theory 23 (2), 127–137.

Chen, L.-S., Kuroda, T., Kusano, T., 1973. Some parabolic equations with unbounded coefficients. Funkcial. Ekvac. 16, 1–28.

Cong, F., Oosterlee, C.W., 2017. Accurate and robust numerical methods for the dynamic portfolio management problem. Comput. Econ. 49 (3), 433–458.

Donnelly, C., Gerrard, R., Guillén, M., Nielsen, J.P., 2015. Less is more: Increasing retirement gains by using an upside terminal wealth constraint. Insurance Math. Econom. 64, 259–267.

Erdélyi, A., 1956. Asymptotic Expansions. In: Dover Books on Mathematics, Dover Publications.

Friedman, A., 1964. Partial Differential Equations of Parabolic Type. Prentice-Hall.

Garlappi, L., Skoulakis, G., 2011. Taylor series approximations to expected utility and optimal portfolio choice. Math. Financ. Econ. 5 (2), 121–156.

Hlawitschka, W., 1994. The empirical nature of taylor-series approximations to expected utility. Amer. Econ. Rev. 84 (3), 713–719.

Jondeau, E., Rockinger, M., 2006. Optimal portfolio allocation under higher moments. Eur. Financial Manag. 12 (1), 29–55.

Karatzas, I., Shreve, S.E., 1998. Brownian Motion and Stochastic Calculus. Springer New York.

Kronborg, M.T., Steffensen, M., 2014. Inconsistent investment and consumption problems. Appl. Math. Optim. 71 (3), 473–515.

Kryger, E., Nordfang, M.-B., Steffensen, M., 2019. Optimal control of an objective functional with non-linearity between the conditional expectations: solutions to a class of time-inconsistent portfolio problems. Math. Methods Oper. Res. 1–34.

Levy, H., Markowitz, H.M., 1979. Approximating expected utility by a function of mean and variance. Amer. Econ. Rev. 308–317.

Loistl, O., 1976. The erroneous approximation of expected utility by means of a taylor's series expansion: Analytic and computational results. Amer. Econ. Rev. 66 (5), 904–910.

Markowitz, H.M., 1952. Portfolio selection. J. Finance 7 (1), 77.

Merton, R.C., 1969. Lifetime portfolio selection under uncertainty: The continuous-time case. Rev. Econ. Stat. 51 (3), 247.

Nordfang, M.-B., Steffensen, M., 2017. Approximations to expected utility optimization in continuous time. SSRN Electron. J..

Porper, F.O., Eidelman, S.D., 1993. Properties of solutions of second-order parabolic equations with lower-order terms. Trans. Mosc. Math. Soc. 1993, 101–137.

Pulley, L.B., 1983. Mean-variance approximations to expected logarithmic utility. Oper. Res. 31 (4), 685–696.

Samuelson, P.A., 1970. The fundamental approximation theorem of portfolio analysis in terms of means, variances and higher moments. Rev. Econom. Stud. 37 (4), 537–542.

Skoulakis, G., 2012. On the quality of taylor approximations to expected utility. Appl. Financial Econ. 22 (11), 863–876.

Steffensen, M., 2004. On merton's problem for life insurers. Astin Bull. 34 (1), 5–25.

Tsiang, S.C., 1972. The rationale of the mean-standard deviation analysis, skewness preference, and the demand for money. Amer. Econ. Rev. 62 (3), 354–371.

Zhang, J., 2005. Representation of solutions to BSDEs associated with a degenerate FSDE. Ann. Appl. Probab. 15 (3), 1798–1831.